
Technical University of Crete
School of Electrical and Computer Engineering
Telecommunication Systems I

Exercise 2

Submission deadline: April 27, 2026

The assignment may be submitted by groups of up to two people

Instructor: Athanasios P. Liavas

Points: 100/1000

On the first page of your report, state the total number of hours required to complete the exercise.

A. In the first part of the exercise, which is mainly experimental, we will study the spectral content of baseband PAM waveforms.

A.1 Construct an SRRC pulse $\phi(t)$ with parameters $T = 10^{-3}$ sec, $\text{over} = 10$, $T_s = \frac{T}{\text{over}}$, $A = 4$, and $a = 0.5$.

(10) Using the functions `fftshift` and `fft`, compute the magnitude of the Fourier transform of $\phi(t)$, $|\Phi(F)|$, at N_f equally spaced points in the interval $[-\frac{F_s}{2}, \frac{F_s}{2}]$.¹ Plot the energy spectral density $|\Phi(F)|^2$ versus the appropriate frequency axis using semilogy.

A.2 Generate a sequence of $N = 100$ independent and equiprobable bits $\{b_0, \dots, b_{N-1}\}$.

Using the mapping

$$0 \longrightarrow +1,$$

$$1 \longrightarrow -1,$$

¹Choose N_f sufficiently large and keep it fixed throughout the exercise. More specifically, N_f should be larger than the length of the waveform $X(t)$ (measured in samples). For example, $N_f = 2048$ is sufficient for relatively small N and over . Otherwise, distortions will appear in the estimated power spectral densities.

map the bits to symbols X_n , for $n = 0, \dots, N - 1$.

Construct the waveform

$$X(t) = \sum_{n=0}^{N-1} X_n \phi(t - nT).$$

Assuming an infinite number of symbols, we proved that the power spectral density of $X(t)$ is

$$S_X(F) = \frac{\sigma_X^2}{T} |\Phi(F)|^2.$$

A.3 (10) Using the functions `fft` and `fftshift`, compute the periodogram of one realization of $X(t)$:

$$P_X(F) = \frac{|\mathcal{F}[X(t)]|^2}{T_{\text{total}}},$$

where T_{total} is the **total time duration** of $X(t)$ in sec. Plot $P_X(F)$ using both plot and semilogy.

Repeat for several different realizations of the bit sequence $\{b_0, \dots, b_{N-1}\}$, so as to get a good sense of what the periodograms of realizations of $X(t)$ look like.

(10) Estimate the power spectral density by computing numerical averages over K (indicatively, $K = 500$) periodogram realizations. On the same semilogy plot, show both the estimate and the theoretical² power spectral density.

(10) As you increase K and N , the approximation should improve. Does this happen in your experiments? If yes, can you explain why?

A.4 Using the mapping

$$00 \longrightarrow +3$$

$$01 \longrightarrow +1$$

$$11 \longrightarrow -1$$

$$10 \longrightarrow -3$$

construct the 4-PAM symbol sequence X_n , for $n = 0, \dots, \frac{N}{2} - 1$. Note that, *if the bits are equiprobable, then the symbols X_n are also equiprobable!*

²Namely, the one obtained from the formula in step A.2, assuming that the energy spectral density of $\phi(t)$ is the $|\Phi(F)|^2$ from step A.1.

Construct the waveform

$$X(t) = \sum_{n=0}^{\frac{N}{2}-1} X_n \phi(t - nT)$$

using the same symbol period T as in question A.2.

(10) Compute the periodogram and estimate the power spectral density by averaging periodogram realizations of $X(t)$. Plot the experimental and the theoretical power spectral density on the same semilogy. What do you observe?

(10) How does the power spectral density of this $X(t)$ compare (in terms of bandwidth and peak magnitude) to the power spectral density of the $X(t)$ in step A.2? Can you explain your observations?

A.5 (10) Repeat step A.3 by setting the symbol period to $T' = 2T$ (keep the sampling period T_s the same as in the previous steps; therefore, you must double the parameter over).

(5) What do you observe about the bandwidth of the waveforms in this case compared to that of the waveforms in step A.3? Can you explain the phenomenon?

A.6 (2.5) If you wanted to transmit data as fast as possible while using the same available bandwidth, would you choose 2-PAM or 4-PAM, and why?

(2.5) If the available bandwidth is very expensive, would you choose symbol period T or $T' = 2T$, and why?

B. In the second part of the exercise, which is mainly theoretical, we will study simple stochastic processes.

Let

$$Y(t) = X \cos(2\pi F_0 t + \Phi), \quad (1)$$

where $X \sim \mathcal{N}(0, 1)$, $\Phi \sim \mathcal{U}[0, 2\pi)$, and X and Φ are independent random variables.

1. (5) Plot, on the same plot, 5 realizations of the process (only for this question, assume $F_0 = 10$ Hz, $T_s = 10^{-2}$ sec, and `t_axis = [0 : T_s : 5/F_0]`).
2. (10) Compute the quantities $\mathcal{E}[Y(t)]$ and $R_{YY}(t + \tau, t) = \mathcal{E}[Y(t + \tau) Y(t)]$. What do you conclude?
3. (5) Compute the power spectral density $S_Y(F)$.